

CASE STUDY 1: HYBRID NETWORK MODERNIZATION, AI DLP & IMMUTABLE RESILIENCE

Enterprise Technical Case Study & Architecture Reference Manual

PRIMARY TECHNICAL DOMAIN

Hybrid SD-WAN, Cloud HA & Backup Resilience

CORE NETWORK PROTOCOLS

BGP (eBGP/iBGP), OSPF, IPsec (IKEv2), VRRP, QoS

ALIGNED CERTIFICATIONS

SC-500, SC-104, Azure Network Engineer, Network+

SECURITY & BACKUP STACK

Purview AI Hub, Front Door, RA-GRS, Defender Apps

I. EXECUTIVE SUMMARY & CONTEXT

This enterprise project modernized a multi-site network infrastructure from high-cost legacy MPLS circuits to a highly resilient hybrid SD-WAN architecture with dynamic eBGP routing. The initiative deployed active-active cloud application availability across Azure Availability Zones behind Azure Front Door, constructed write-once immutable storage vaults (RA-GRS) for ransomware resilience, and enforced Microsoft Purview AI security controls to eliminate sensitive data exposure to public LLMs while achieving 99.99% operational uptime.

II. STAKEHOLDER DISCOVERY & INTERVIEW MATRIX

Prior to engineering design, structured discovery interviews were conducted across leadership roles to map technical pain points, budget risks, and compliance requirements.

Stakeholder Role	Key Concerns & Questions	Uncovered Pain Points	Design Resolution
CISO / Chief Security Officer	- How are employees using public GenAI tools? - What safeguards prevent code leaks into LLMs?	Zero knowledge of Shadow AI usage. - Unrestricted paste actions into prompt boxes.	Designed Microsoft Purview AI DLP rules with custom SITs & Defender session controls.
VP of Infrastructure	- Why do branch sites experience packet loss? - What is the cost impact of legacy MPLS?	- Single-homed MPLS (\$45k/mo) with brownouts. - HQ hairpinning causing 85ms+ latency.	Re-architected WAN to Hybrid SD-WAN with eBGP over broadband & ExpressRoute.
Director of Compliance & Risk	- How resilient are backups to ransomware? - Can we satisfy SOC 2 & HIPAA storage rules?	- On-prem backups reachable on flat subnet. - Lack of write-once immutable storage.	Built Azure RA-GRS Immutable Storage with Multi-User Authorization & PITR.
Lead Cloud Architect	- What if an Azure region experiences an outage? - Can updates deploy without downtime?	- Single Availability Zone hosting. - Updates required 2-hour maintenance windows.	Architected multi-AZ design behind Azure Front Door with blue/green deployment.

III. FORMAL SCOPE BOUNDARY STATEMENT

IN-SCOPE Deliverables • Hybrid Network Edge: Dual SD-WAN appliances across 12 branch sites with eBGP/OSPF dynamic routing and ExpressRoute failover. • AI Data Governance: Microsoft Purview AI Hub, SC-500/SC-104 DLP policies, and Defender for Cloud Apps proxy controls. • High Availability & DR: Multi-AZ cloud web hosting, Azure Front Door, and RA-GRS immutable backup vaults (RPO < 15m, RTO < 35m). • Operational Schedule: All cutovers restricted to non-business hours maintenance windows.	OUT-OF-SCOPE Boundaries • Physical recabling or replacement of internal branch Layer-2 access switches. • Full application code refactoring of legacy monolithic on-premises ERP databases. • End-user desktop hardware procurement and local operating system provisioning. • Migration of external customer identity directories (kept isolated).
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IV. PRE-DESIGN BASELINE AUDIT ASSESSMENT

88 ms AVG CLOUD LATENCY (MPLS HAIRPIN)	310 / day UNMONITORED GENAI CODE INTERCEPTS	4.5 Hours LEGACY DR RESTORE WINDOW (RTO)
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V. CHRONOLOGICAL STEP-BY-STEP EXECUTION SEQUENCE

PHASE 1	Pre-Design Audit: Captured baseline telemetry across 12 branches and conducted Purview scans to address zero knowledge of GenAI shadow usage.
PHASE 2	Underlay & ExpressRoute Provisioning: Installed redundant ExpressRoute circuits and established eBGP dynamic peering with internal OSPF distribution.
PHASE 3	SD-WAN Edge & IPsec Overlay: Deployed dual SD-WAN appliances per site, configuring IPsec (IKEv2) backup tunnels over broadband with local VRRP redundancy.
PHASE 4	Multi-AZ Cloud Hosting: Migrated workloads to Azure Availability Zones behind Azure Application Gateway WAF v2 and Azure Front Door global routing.
PHASE 5	Immutable Backup Vaults: Constructed Azure Backup vaults with Read-Access Geo-Redundant Storage (RA-GRS) and Write-Once, Read-Many (WORM) policies.
PHASE 6	SC-500/SC-104 Purview DLP: Enforced custom Sensitive Info Types (SITs) for financial records and source code; routed LLM prompts through Defender for Cloud Apps.
PHASE 7	Non-Business Cutover: Executed final WAN and DNS traffic shifts during scheduled non-business hours windows, validating seamless VRRP/BGP failover.
PHASE 8	Validation & DR Testing: Simulated zone outages and ransomware deletion attempts, verifying sub-35-minute RTO and 100% AI DLP prompt interception.

■ VI. KEY METRICS & TECHNICAL IMPACT SUMMARY

Metric Category	Pre-Implementation Baseline	Post-Implementation Result
Network Uptime	99.85% (Frequent MPLS circuit brownouts)	99.99% Availability via automated BGP failover
Monthly WAN Expense	\$45,000 / month across 12 branch sites	38% Cost Reduction (\$27,900 / month total)
AI Prompt DLP Exposure	Unmonitored / 310 daily risk events	1,400+ Exfiltration attempts blocked (100% compliance)
Disaster Recovery RTO	4.5 Hours (Manual DNS failover process)	< 35 Minutes via automated recovery runbooks

■ VII. TECHNICAL ACCOMPLISHMENT SUMMARY

- **Hybrid Network Resilience:** Engineered dual-hub hybrid SD-WAN infrastructure with eBGP/OSPF dynamic routing and ExpressRoute/IPsec failover across 12 sites, achieving 99.99% multi-site availability and reducing monthly WAN expenditures by 38%.
- **Cloud Architecture & Immutable DR:** Architected multi-region active-active cloud applications across Azure Availability Zones backed by Azure Front Door and RA-GRS immutable backups (WORM), establishing an RPO < 15 minutes and RTO < 35 minutes.
- **AI Security & Data Protection:** Implemented Microsoft Purview DLP policies targeting GenAI prompts, securing enterprise financial records and proprietary source code against unauthorized uploads with 100% policy enforcement (SC-500 / SC-104 compliant).

■ VIII. STAKEHOLDER SENTIMENT, FINANCIAL MODEL & PROJECT OUTCOME

Financial Transparency & Project Transformation Success

During initial discovery, executive stakeholders expressed significant hesitation regarding the project, driven by concerns over the heavy capital expenditure (CapEx) typically required for enterprise network modernization and cloud security overhaul.

We eliminated these concerns by presenting a lean operational model (OpEx) that removed capital overhead entirely. We demonstrated that the financial commitment was limited strictly to:

- **Implementation Team Fees:** Flat-rate professional services for architecture design and deployment.
- **Maintenance & Monitoring Fees:** Managed operational support for network uptime and security telemetry.
- **Azure Cost Management Utility:** Implementation of the Azure utility to provide continuous visibility, budget alerting, and monthly cost optimization across cloud resources.

Reassured by the predictable operational expenses and lack of capital investment, stakeholders signed the contract and authorized immediate work. The project was completed as a major success:

- **Eliminated Bottlenecks:** All network bottlenecking was eliminated across branches and cloud endpoints using BGP path selection and ExpressRoute peering.
- **File Lifespan Tagging:** Implemented a redundant backup framework with metadata tags assigning a defined lifespan to files for automated lifecycle management and purging.
- **High Stakeholder Confidence:** Stakeholders reported feeling **more secure than they had in the last 10 years**, citing full visibility into AI interactions, rock-solid network availability, and ransomware-proof data immutability.